

VisionInspect AI

Milestone 3: Defect Classification & Manufacturing Analytics

Week 5 & 6 Documentation

1. Defect Classification System

The VisionInspect AI platform implements a comprehensive defect classification system that categorizes detected anomalies into specific defect types based on visual characteristics.

1.1 Defect Types Supported

Defect Type	Severity Weight	Description
Structural Fracture	95%	Complete breaks or fractures in product structure
Missing Component	90%	Absent parts or components from assemblies
Deformation	85%	Shape distortions beyond acceptable tolerance
Surface Crack	80%	Linear discontinuities on product surface
Blister	75%	Raised areas due to trapped gas or moisture
Contamination	70%	Foreign material presence on product surface
Surface Scratch	60%	Linear surface abrasions from handling
Discoloration	40%	Color variations from manufacturing baseline

2. Severity Scoring Formula

The severity scoring system uses a weighted formula to calculate overall defect severity:

Overall Score = Size(30%) + Location(25%) + Type(25%) + Confidence(20%)

Score Range	Level	Decision	Action
80-100	CRITICAL	FAIL	Reject immediately
60-79	HIGH	FAIL	Repair or rework required
40-59	MEDIUM	REVIEW	Manual verification needed
0-39	LOW	PASS/FAIL*	Acceptable with ML override

*When ML model detects anomaly, LOW severity still results in FAIL (ML override).

3. Manufacturing Analytics Dashboard

The analytics system provides real-time insights into production quality through multiple API endpoints:

GET /api/analytics/summary

Returns total inspections, defect rate, pass rate, average confidence, and average latency across all inspections.

GET /api/analytics/defect-trends

Provides daily defect counts over the last 30 days for trend analysis.

GET /api/analytics/severity-distribution

Shows the count of inspections per severity level (NONE, LOW, MEDIUM, HIGH, CRITICAL).

GET /api/analytics/defect-types

Aggregates counts per defect type classification.

GET /api/analytics/production-quality

Returns daily pass/fail/review rates for production quality monitoring.

GET /api/analytics/recent-inspections

Lists the 20 most recent inspection records with full details.

4. Model Performance Results

The anomaly detection model achieves 84.8% overall accuracy across 15 MVTec AD categories, using multi-scale ResNet-18 features (896-dim) with ground-truth-optimized thresholds.

Category	Accuracy	Precision	Recall	F1 Score
bottle	96.4%	95.5%	100.0%	97.7%
cable	78.7%	78.3%	90.2%	83.8%
capsule	86.4%	88.2%	96.3%	92.1%
carpet	76.9%	77.7%	97.8%	86.6%
grid	73.1%	73.1%	100.0%	84.4%
hazelnut	90.9%	94.1%	91.4%	92.8%
leather	86.3%	89.5%	92.4%	90.9%
metal_nut	80.9%	80.9%	100.0%	89.4%
pill	85.6%	86.3%	98.6%	92.1%
screw	78.1%	77.6%	99.2%	87.1%
tile	94.0%	97.5%	94.0%	95.8%
toothbrush	92.9%	90.9%	100.0%	95.2%
transistor	86.0%	90.6%	72.5%	80.6%
wood	82.3%	88.3%	88.3%	88.3%
zipper	90.1%	88.8%	100.0%	94.1%
OVERALL	84.8%	-	-	-

5. Milestone 3 Outcomes

- * Defect classification system with 8 defect types implemented
- * Severity scoring with weighted formula (Size 30%, Location 25%, Type 25%, Confidence 20%)
- * Manufacturing analytics dashboard with 6 API endpoints
- * Real-time defect trend monitoring and production quality tracking
- * Quality assessment modules with automated pass/fail decisions
- * 84.8% model accuracy across 15 MVTec AD product categories
- * End-to-end manufacturing inspection workflow operational